

Q How to find unit digits of various number formats?

24/7/25

Q Find the

family of last digit / unit digit.

0, 1, 5, 6	2, 3, 7, 8	4, 9
↓ ↓ ↓ ↓		
0, 1, 5, 6	cycle = 4	odd = 4
(leaves last digit itself only)		even = 6
		odd = 9
		even = 1

$$2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 32$$

repeat after 4 cycle

Question

Q Find the unit digit of

$$(2153)^{167} \times (8267)^{153}$$

Solⁿ

$$(3)^{167} \times (7)^{153}$$

divid $\rightarrow \frac{\text{pow}}{4}$

Remainder = 0

Remainder $\neq 0$ (Something)

$$\begin{array}{r} 167 \\ 4 \overline{) 167} \\ \underline{160} \\ 07 \\ 4 \overline{) 07} \\ \underline{4} \\ 3 \end{array}$$

for 3

$$\begin{array}{r} 153 \\ 4 \overline{) 153} \\ \underline{120} \\ 33 \\ 4 \overline{) 33} \\ \underline{32} \\ 1 \end{array}$$

for 7

for 5×2 (5) ²⁸

$$\begin{array}{r} 4 \overline{) 48} \quad 12 \\ \underline{4} \\ 08 \\ \underline{8} \\ 0 \end{array}$$

(2) $\checkmark$

Sol²

(3)¹⁶⁷ $\times$ (7)¹⁵³

(3)³ $\times$ (7)¹

$$\begin{array}{r} 27 \times 7 \\ \hline 189 \end{array}$$

Ans = 9

② Find the unit digit of

(432)⁴¹² $\times$ (499)⁴³¹

Sol²

= (2)⁴¹² $\times$ (9)⁴³¹

= (2)⁴ $\times$ (9)^{odd}

→ gives 9

= (2)⁴ $\times$ (9)

= 16 $\times$ 9

= 144

Ans = 4

$$\begin{array}{r} 4 \overline{) 412} \quad (103) \\ \underline{4} \\ 012 \\ \underline{12} \\ 00 \end{array}$$

5 6 remainder

③ Find unit digit of

$$(217)^{413} \times (819)^{547} \times (414)^{624} \times (342)^8$$

$$= (7)^{413} \times (9)^{547} \times (4)^{624} \times (2)^8$$

$$\begin{array}{l} 4 \overline{) 413} 103 \\ \underline{4} \\ 013 \\ \underline{12} \\ 1 \end{array} \quad \left| \begin{array}{l} = 9^{\text{odd}} \\ = 9 \end{array} \right. \quad \left| \begin{array}{l} = 4^{\text{even}} \\ = 6 \end{array} \right. \quad \left| \begin{array}{l} 4 \overline{) 8} (2) \\ \underline{8} \\ 0 \end{array} \right.$$

$$= (7)^1 \times (9) \times (6) \times (2)^4$$

$$= 7 \times 9 \times 6 \times 16$$

$$= 63 \times 18 \times 16$$

$$= 63 \times 6 \times 16$$

$$= 18 \times 16$$

$$= 48$$

Ans = 8

$$\textcircled{Q} \text{ if } X = (164)^{169} + (333)^{337} - (727)^{726}$$

Then what is the unit digit of X ?

Solⁿ

$$X = (4)^{169} + (3)^{337} - (7)^{726}$$

$= 4^{\text{odd}}$

$= -4$

$$\begin{array}{r} 4 \overline{) 337} \quad (84) \\ \underline{32} \\ 017 \\ \underline{16} \\ 01 \end{array}$$

$$\begin{array}{r} 4 \overline{) 726} \quad (181) \\ \underline{4} \\ 32 \\ \underline{32} \\ 006 \\ \underline{4} \\ 2 \end{array}$$

$$X = 4 + (3)^1 - (7)^2$$

$$X = 4 + 3 - 49$$

$$X = 7 - 49$$

Ans $\rightarrow$ ~~7~~ ~~49~~

$$X = 7 - 9$$

$$\boxed{X = -2}$$

Add

$$X = -2 + 10$$

$$\boxed{X = 8}$$

(for -ve ~~if~~ add 10)

⑤ Find the unit digit of $((137)^{13})^{47}$

Solⁿ

Given

$$((137)^{13})^{47}$$

W.K.T

$$(a^m)^n = a^{m \times n}$$

$$\text{So } = (137)^{13 \times 47}$$

$$= (7)^{\frac{13}{4} \times \frac{47}{4}}$$

$$= (7)^{1 \times 3}$$

$$= (7)^3 = 343$$

$$\begin{array}{r} 4 \overline{) 1373} \\ \underline{12} \\ 07 \end{array}$$

$$\begin{array}{r} 4 \overline{) 4711} \\ \underline{4} \\ 07 \\ \underline{4} \\ 03 \end{array}$$

$$\begin{array}{r} 49 \\ \times 7 \\ \hline 343 \end{array}$$

Ans = 3

⑥ If $N = 7^{95} - 3^{58}$, then the digit at the unit place of N is !

Solⁿ

$$= (7)^{95} - (3)^{58}$$

$$= (7)^3 - (3)^2$$

$$= 343 - 9$$

$$= 3 - 9$$

$$= -6$$

$$= -6 + 10$$

$$\begin{array}{r} 4 \overline{) 9523} \\ \underline{8} \\ 15 \\ \underline{12} \\ 3 \end{array}$$

$$\begin{array}{r} 4 \overline{) 5814} \\ \underline{4} \\ 18 \\ \underline{16} \\ 2 \end{array}$$

Ans = 4

⑦ Find the unit digit of the product of all the prime no's

Solⁿ Prime no's it should be 2 factory

$$2 < \frac{1}{2}$$

$$3 < \frac{1}{3}$$

$$5 < \frac{1}{5}$$

* the no which is divisible by 1 & itself only.

* 1 is not a prime no & composite.

(Composite no's) = not prime no's

w.k.t

$$\text{Prime} = 2 \times 3 \times 5 \times 7$$

$$\underbrace{2 \times 3}_{6} \times 5 \times 7$$

$$3 \times 0$$

When the 0 is multiple by all the unit digit of prime number is 0

Ans $\boxed{0}$

⑧ And the last digit of $(32)^{32 \times 32}$,
then the digit at the unit place of
N is:

Solⁿ

$$= (32)^{32 \times 32}$$

$$= (2)^{\frac{32}{4} \times \frac{32}{4}}$$

$$= (2)^{4 \times 4}$$

$$= (2)^{16}$$

$$= (2)^4 = 16$$

$$\begin{array}{r} 4 \overline{) 32} \quad (8 \\ 32 \\ \hline 00 \end{array}$$

$$\begin{array}{r} 4 \overline{) 16} \quad (4 \\ 16 \\ \hline 00 \end{array}$$

Ans 16